

- 4 (a) e.g. no energy transfer
amplitude varies along its length/nodes and antinodes
neighbouring points (in inter-nodal loop) vibrate in phase, etc.
(any two, 1 mark each to max 2B2 [2]

Page 3	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE AS/A LEVEL – May/June 2010	9702	22

- (b) (i) $\lambda = (330 \times 10^2)/550$ M1
 $\lambda = 60 \text{ cm}$ A0 [1]
- (ii) node labelled at piston B1
 antinode labelled at open end of tube B1
 additional node and antinode in correct positions along tube B1 [3]
- (c) at lowest frequency, length = $\lambda/4$ C1
 $\lambda = 1.8 \text{ m}$
 frequency = $330/1.8$ C1
 $= 180 \text{ Hz}$ A1 [3]

5 (a) (i) Young modulus = stress/strain

C1